

Cloud Computing

Objectives

Introduction to cloud computing

Basic concepts and Terminology

Goals and Benefits

Risks and Challenges

Brief History

John McCarthy publicly proposed in 1961 If computers of the kind I have advocated become the computers of the future, then computing may someday be organized as a public utility just as the telephone system is a public utility. ... The computer utility could become the basis of a new and important industry.”

In 1969, Leonard Kleinrock a chief scientist of the Advanced Research Projects Agency Network or ARPANET project that seeded the Internet, stated: “As of now, computer networks are still in their infancy, but as they grow up and become sophisticated, we will probably see the spread of ‘computer utilities’ ...”.

Brief History

The general public has been leveraging forms of Internet-based computer utilities since the mid-1990s through various incarnations of search engines (Yahoo!, Google), e-mail services (Hotmail, Gmail), open publishing platforms (MySpace, Facebook, YouTube), and other types of social media (Twitter, LinkedIn). Though consumer-centric, these services popularized and validated core concepts that form the basis of modern-day cloud computing

In the late 1990s, Salesforce.com pioneered the notion of bringing remotely provisioned services into the enterprise.

In 2002, Amazon.com launched the Amazon Web Services (AWS) platform, a suite of enterprise-oriented services that provide remotely provisioned storage, computing resources, and business functionality.

Brief History

It wasn't until 2006 that the term “cloud computing” emerged in the commercial arena.

It was during this time that Amazon launched its Elastic Compute Cloud (EC2) services that enabled organizations to “lease” computing capacity and processing power to run their enterprise applications.

Google Apps also began providing browser-based enterprise applications in the same year, and three years later, the Google App Engine became another historic milestone.

Definition of Cloud Computing

The U.S. National Institute of Standards and Technology (NIST) defines cloud computing as:

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model is composed of **five essential characteristics**, **three service models**, and **four deployment models**

الصفحة السادسة: يتعرض تعريف المعهد الوطني الأمريكي للمعايير والتكنولوجيا (NIST) التعريف بيقول إنها نموذج يسمح بوصول سهل ومريح وفي أي وقت عبر الشبكة. (NIST) لمجموعة من موارد الحوسبة المشتركة (زي الشبكات، السيرفرات، التخزين، والتطبيقات). الموارد دي ممكن توفيرها وإطلاقها بسرعة وبأقل مجهود في الإدارة أو تفاعل مع مقدم الخدمة. النموذج ده بيتكون من 5 خصائص أساسية، 3 نماذج للخدمات، و 4 نماذج للانتشار والتوزيع

More concise Definition

This book provides a more concise definition:

“Cloud computing is a specialized form of **distributed computing** that introduces utilization models for remotely provisioning scalable and measured resources.”

This simplified definition is in line with all of the preceding definition variations that were put forth by other organizations within the cloud computing industry

التعريف ده بيوفر نماذج استخدام بتسمح بتوفير موارد عن بُعد بتكون قابلة للتوسع ومحسوبة بدقة. التعريف المبسط ده بيتماشى تماماً مع التعريفات الثانية الموجودة في صناعة الحوسبة السحابية

Business Drivers

The primary business drivers that exposed the need for cloud computing and led to its formation include

- capacity planning

- Cost reduction

- and organizational agility.

Capacity Planning

Capacity planning

Process of determining and fulfilling future demand for IT resources

Capacity vs. demand

Discrepancy between capacity of IT resources and actual demand

Over-provisioning: resource capacity exceeds demand

Under-provisioning: demand exceeds resource capacity

Capacity planning aims to minimize the discrepancy of available resources vs. demand

Capacity Planning Strategies

Lead Strategy – adding capacity to an IT resource in anticipation of demand

Lag Strategy – adding capacity when the IT resource reaches its full capacity

Match Strategy – adding IT resource capacity in small increments, as demand increases

Challenges

Planning for capacity can be challenging because it requires estimating usage load fluctuations. **تقلبات الاستخدام**

There is a constant need to balance peak usage requirements without unnecessary over-expenditure on infrastructure.

An example is outfitting IT infrastructure to accommodate maximum usage loads which can impose unreasonable financial investments.

In such cases, moderating investments can result in under-provisioning, leading to transaction losses and other usage limitations from lowered usage thresholds.

بنتاقش التحديات، زي إن تخطيط السعة صعب لأنه بيحتاج توقع لتقلبات الاستخدام.
في حاجة مستمرة للموازنة بين توفير الموارد لأوقات الذروة، وبين عدم صرف فلوس كتير على البنية التحتية بدون داعي
لو جهزت البنية التحتية عشان تستحمل أقصى حمل ممكن، ده هيكلفك.
استثمارات مالية ضخمة جدا
ولو قللت الاستثمارات أكثر من اللازم، ده هيودي لنقص في الموارد،
وبالتالي خسارة في المعاملات وفرض قيود على الاستخدام

Cost Reduction

A direct alignment between IT costs and business performance can be difficult to maintain.

The growth of IT environments often corresponds to the assessment of their maximum usage requirements.

This can make the support of new and expanded business automations an ever-increasing investment.

Two costs need to be accounted for: the cost of acquiring new infrastructure, and the cost of its ongoing ownership.

Operational overhead represents a considerable share of IT budgets, often exceeding up-front investment costs.

Cost Reduction

Cost reduction

- IT Infrastructure **acquisition**
- IT Infrastructure **maintenance**

Operational overhead

- Technical personnel to maintain physical IT infrastructure
- System upgrades, patches that add testing to deployment cycles
- Utility bills, capital investments for power and cooling
- Security and access control measures for server rooms
- Admin and accounting staff to track licenses, support agreements, purchases

Organizational Agility

Organizational agility is the measure of an organization's responsiveness to change.

An IT enterprise often needs to respond to business change by scaling its IT resources beyond the scope of what was previously predicted or planned for.

For example, infrastructure may be subject to limitations that prevent the organization from responding to usage fluctuations

Technology Innovations

The primary technology innovations that influenced and inspired key distinguishing features and aspects of cloud computing include -

- Clustering
- grid computing
- traditional forms of virtualization